Module-I "Basic Calculus":-

Improper Integrals Type-I:

→ When the limit are infinite.

→ $$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty}\int_a^{t} f(x)\,dx$$

→ $$\int_{-\infty}^{b} f(x)\,dx = \lim_{t\to-\infty}\int_t^{b} f(x)\,dx$$

Improper Integrals Type-II:-

→ $\int_a^b f(x)\,dx$, where $f(x)\to\infty$ at $x=c$

→ $$\int_a^{c} f(x)\,dx = \lim_{t\to c^-}\int_a^{t} f(x)\,dx$$

→ $$\int_c^{b} f(x)\,dx = \lim_{t\to c^+}\int_t^{b} f(x)\,dx$$

Gamma function →

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x}\,dx$$

→ Properities of Gamma function:-

1) $\Gamma(n+1) = n\Gamma n$

2) $\Gamma n = (n-1)!$

3) $\Gamma \tfrac{1}{2} = \sqrt{\pi}$

Beta Function :-

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$$

→ Properties of Beta function :-

$$\beta(m,n) = B(n,m)$$

$$\beta(m,n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma m+n}$$

Volume of Revolution :-

About x axis $\Rightarrow$ $V = \pi \int_a^b [f(x)]^2\, dx$

About y axis $\Rightarrow$ $V = \pi \int_a^b [f(y)]^2\, dy$

Shell method $\Rightarrow$ $V = 2\pi \int_a^b x \cdot f(x)\, dx$

Surface area : of Revolution :-

About x axis $= S = 2\pi \int_a^b f(x)\sqrt{1+(f'(x))^2}\, dx$

About y axis $= S = 2\pi \int_a^b x \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$

odule: 2 "Single Variable Calculus (Differentiation)":-

Taylor Series :-

$$F(x) = F(a) + F'(a)(x-a) + \frac{F''(a)}{2!}(x-a)^2 + \ldots \ldots + \frac{F^n(a)}{n!}(x-a)^n + \ldots$$

Maclaurin Series :-

$$F(x) = F(0) + F'(0).x + \frac{F''(0)}{2!}x^2 + \ldots + \frac{F^n(0)}{n!}.x^n + \ldots$$

→ Maclaurin Series is the Special Example of Taylor Series when $(a=0)$.

All indeterment forms :-

1) $0/0$

2) ∞/∞

3) $0.\infty$

4) $\infty - \infty$

5) 0^0

6) ∞^0

7) 1^∞

L'Hopital's Rule :-

If, $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$ or $\pm \infty$

and the derivatives $f'(x)$ and $g'(x)$ exist near a, and $g'(x) \neq 0$ then,

$$\boxed{\lim_{a \to \infty} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}}$$ If limit on RHS exist.

it is Valid only indeterminant form $\frac{0}{0}$ or $\frac{\infty}{\infty}$!

and, other all Indetenminant forms convert in this two indetenminant form.

Module-3 "Sequence and Series":

Defination of Sequence is convergent or divergent

$\lim_{n\to\infty} u_n = l$ $\quad l$ = finite / unique $\Rightarrow$ converges
$l = \infty / -\infty$ $\Rightarrow$ diverges
l = non unique $\Rightarrow$ oscillatory

Sandwich theorem:-

$\{a_n\}\ \{b_n\}\ \{c_n\}$

$a_n \leq b_n \leq c_n$

If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = l$ $\quad \boxed{\therefore \lim_{n\to\infty} b_n = l}$

Geometric Series:-

$= a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$

$= \sum_{n=1}^{\infty} ar^{n-1}$

$\Rightarrow$ Conditions: $|r| < 1 \Rightarrow$ convergent.

$\boxed{\frac{a}{1-r}}$, at convergence

Telescoping Series:-

Ex: $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

Partial Fraction:

$\frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{(n+1)}$ and find A and B and evaluate this Telescoping Series.

n^{th} Term Test :-

$\lim_{n\to\infty} a_n = l \Rightarrow l \neq 0$ The Sequance is always diverges.

$\Rightarrow l = 0$ No conclusun.

General Properties of Series :-

(1) $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} v_n =$ Converges.

$\therefore \sum_{n=1}^{\infty} (u_n + v_n) = \sum_{n=1}^{\infty} (u_n - v_n) -$ Convergence.

(2) $\sum_{n=1}^{\infty} u_n =$ convergence & $\sum_{n=1}^{\infty} v_n =$ divergence.

$\therefore \sum_{n=1}^{\infty} (u_n + v_n) = \sum_{n=1}^{\infty} (u_n - v_n) =$ diverge

Note :- If a (+ve) term Series $\sum u_n$ is Convergent then $\lim_{n\to\infty} u_n = 0$, $\lim_{n\to\infty}$ is necesriey condition. But not sufficient condition for convergent of Series.

Integral tests :-

→ It applicable only Positive term Series.

$f(1) + f(2) + f(3) + \ldots + f(n) + \ldots$ where; $f(n)$ decreases as n increses converge or diverge accroding as,

$\longrightarrow \lim_{b\to\infty} \int_1^b f(x)\,dx$ is finite or infinit

P-series Test:

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ $p \neq -1$. P is a real Constant

$p > 1$ then Series is Convergent.

$p \leq 1$ then Series is divergent.

The Integral test is most effective when the function f to be used is easily integrated.

Direct Comparison :-

test $\sum a_n$:

→ Compare with a known Positive Series $\sum b_n$

→ If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges then, $\sum a_n$ also converges.

→ If $a_n \geq b_n \geq 0$ and $\sum b_n$ diverges then, $\sum a_n$ also diverges.

limit Comparison :-

for Series $\sum a_n$ and $\sum b_n$. (Positive term)

$L = \lim_{n \to \infty} \frac{a_n}{b_n}$ If L is a positive finite number. $(0 < L < \infty)$

→ If The Ratio is Constant, They behave the Same

Ratio Test :-

$$L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

- If $L < 1$ → Series convergent
- If $L > 1$ → Series divergent
- If $L = 1$ → test fails.

Root test :-

$$L = \lim_{n \to \infty} \sqrt[n]{|a_n|}$$

- If $L < 1$ → Convergent
- If $L > 1$ → divergent
- If $L = 1$ → test fails.

Libnitz Test :-

→ Alternate terms (+ve) or (-ve),
→ decreasing series.

$a_1 > a_2 > a_3, \ldots$

$\lim_{n \to \infty} u_n = 0 \Rightarrow$ convergent

$\lim_{n \to \infty} u_n \neq 0 \Rightarrow$ oscillatory.

Absolute Convergent Test:

If $\sum_{n=1}^{\infty} |a_n|$ converges then $\sum_{n=1}^{\infty} a_n$ is Converges

Conditionaly Convergent:

A Series that Convergent but does not converge absoluty converges. is conditionaly convergent.

Power Series:

A Power series is an infinite series of the form =

$$\boxed{\sum_{n=0}^{\infty} a_n (x-c)^n}$$

a_n = coefficient
c = Center
It behaves like a polynomial of Infinite degree.

Radius of Convergence :- (R):

It tells how far from the center c the Series converges.

Interval of Convergence :-

once you find (R)., The interval is,

$(c-R, c+R)$ → The check the end points $x = c-R$ and $x = c+R$ Sepratly to see if they converge or diverge.

Module-4 "Multivariable Calculus" (Differentiation)

(1) Limit :-

$$\boxed{\lim_{(x,y)\to(a,b)} f(x,y)}$$

(2) Continuity :-

A function $f(x,y)$ is continuous at (a,b) if

$$\boxed{\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)}$$

(3) Differentiability :-

$f(x,y)$ is differentiable at (a,b) if,

$$\boxed{f(a+h, b+k) = f(a,b) + f_x(a,b)h + f_y(a,b)k + o(\sqrt{h^2+k^2})}$$

Were, the last term becomes very small compared to $\sqrt{h^2+k^2}$.

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1) Limit (n variables) :

For $f(x_1, x_2, \dots, x_n)$ at $(a_1, a_2, \dots, a_n)$:

$$\lim_{x\to a} f(x).$$

Exists if the value is the same along every path in n-dimentional space.

(2) Continuity :-

→ f is continuous at $a = (a_1, \dots, a_n)$ if:-

$$\boxed{\lim_{x \to a} f(x) = f(a)}$$

(3) Differentiability:

f is differentiable at a if :-

$$\boxed{f(a+h) = f(a) + \nabla f(a) \cdot h + o(\|h\|)}$$

Where :- • $\nabla f(a)$ = gradiant
• $h = (h_1, h_2, h_3, \dots)$
• $o(\|h\|)$ = very small error term.

Gradient (∇f) :-

→ for a function of n variables :-

$$f(x_1, x_2, x_3, \dots, x_n).$$

→ The gradiant is the vector of all first partial derivatives :

$$\boxed{\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3}, \dots, \frac{\partial f}{\partial x_n}\right)}$$

Directional Derivative :-

→ for a function $f(x_1, x_2, x_3, \dots x_n)$ the directional derivative at point a in the direction of a unit vector u is,

$$\boxed{D_u f(a) = \nabla f(a) \cdot u}$$

Direction derivative = gradient . direction

Tangent Plane and normal line to the Surface $f(x, y, z) = C$;

(1) Tangent Plane :-

$$\boxed{(x - x_0) f_{x(a_0)} + (y - y_0) f_{y(a_0)} + (z - z_0) f_{z(a_0)} = 0}$$

were, $a_0 = (x_0, y_0, z_0)$

(2) Normal Line :-

$x = x_0 + f_x(x_0, y_0, z_0)\, t$
$y = y_0 + f_y(x_0, y_0, z_0)\, t$
$z = z_0 + f_z(x_0, y_0, z_0)\, t$

Extream Value of function of two Variable

(1) Critical Point:

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

Solve this to get point (x_0, y_0)

(2) second derivative:-

$$f_{xx}, \quad f_{yy} \quad f_{xy}.$$

then find.

$$D = f_{xx}(x_0, y_0) \cdot f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

(3) classification:

(1) Local minimum: $D > 0$ and $f_{xx} > 0$

(2) Local Maximum: $D > 0$ and $f_{xx} < 0$

(3) Saddle point: $D < 0$

(4) Test fails: $D = 0$ No Conclusun.

Method of Lagrange Multiples :

Lagrange Multiples :

use to find maximum / minima of $f(x,y)$ subject to a constant

$$\boxed{g(x,y) = c}$$

Main Condition :

$$\boxed{\nabla f = \lambda \nabla g}$$

- $\nabla f = (f_x, f_y)$
- $\nabla g = (g_x, g_y)$
- λ = Lagrange multiples.

System of equation :-

$$f_x = \lambda g_x$$

$$f_y = \lambda g_y \qquad g(x,y) = c$$

This gives Candidates Point for maxima / minima

For 3 Variables :

constraint : $g(x,y,z) = c$

$$f_x = \lambda g_x$$

$$f_y = \lambda g_y$$

$$f_z = \lambda g_z \qquad g(x,y,z) = c$$

Module-5 "Multiple Variable Calculus" (Integration).

(1) Double Integration (Cartesian Coordinates)

For a region R,

$$\iint_R f(x,y)\, dA$$

→ <u>Type-I</u> <u>Region</u> (x constant to x, y varies):

$$\boxed{\iint_R f(x,y)\, dA = \int_{x=a}^{b} \int_{y=g_1(x)}^{g_2(x)} f(x,y)\, dy\, dx}$$

→ Type-II Region (y = constant to y, x varies):

$$\boxed{\iint_R f(x,y)\cdot dA = \int_{y=a}^{b} \int_{y=g_1(y)}^{g_2(y)} f(x y)\, dx\, dy}$$

(2) Double Integration (Polar Co-ordinates):

→ <u>Use Polar Substitution:</u>

$$x = r\cos\theta \quad , \quad y = r\sin\theta$$

Area element : $dA = r\,dr\,d\theta$

→ Polar Double integration:

$$\boxed{\iint_R f(x,y)\, dA = \int_{\theta=\alpha}^{\beta} \int_{r=a(\theta)}^{b(\theta)} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta}$$

Change of Order of Integration :-

To change the order of Integration in a double integral, you rewrite the region of integration. R with reversed limit.

- Orignal form :-

$$\int_{0}^{b}\int_{g_1(x)}^{g_2(x)} f(x,y)\, dy\, dx$$

- Reversed form :-

$$\int_{c}^{d}\int_{h_1(y)}^{h_2(y)} f(x,y)\, dx\, dy$$

Steps :-

(1) Draw/visualize the region R.

(2) Identify the x-range and y range.

(3) Rewrite the Same region with y as outer and x as inner variable.

(4) Replace limit accordingly.

Applications of Double Integrals :-

(1) Area of Region :- $\boxed{A = \iint_R 1 \cdot dA}$

(2) Volume under a Surface : $\boxed{V = \iint_R f(x,y)\, dA}$

(3) Center of mass (constant density):

$$\boxed{\bar{x} = \frac{1}{A}\iint_R x\,dA} \quad \boxed{\bar{y} = \frac{1}{A}\iint_R y\,dA}$$

(4) Moments:

$$\boxed{M_x = \iint_R y\rho\,dA} \quad \boxed{M_y = \iint_R x\rho\,dA}$$

Triple Integrals -

(1) Cartesian,

$$\boxed{\iiint_D f(x,y,z)\,dV}$$

Volume element :- $\boxed{dV = dx\,dy\,dz}$

(2) Cylindrical Co-ordinates :-

$x = r\cos\theta$, $y = r\sin\theta$ $z = z$

Volume element :- $dV = r\,dr\,d\theta\,dz$

Integral :-

$$\boxed{\iiint_D f(r,\theta,z)\,r\,dr\,d\theta\,dz}$$

(3) Sphirical Co-ordinates :-

$x = \rho \sin\phi \cos\theta$

$y = \rho \sin\phi \sin\theta$

$z = \rho \cos\phi$

Volume element : $\boxed{dv = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta}$

- Integral :

$$\boxed{\iiint_{D} f(\rho, \phi, \theta) \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta}$$